

TPL-11 · FREE TEMPLATES

Quantitative schedule risk analysis input sheet

The inputs a simulation runs on, elicited so that the answer means something

VERSION 1.0 · DRAFT
2026-08-04

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— Summary

The three sheets a quantitative schedule risk analysis actually needs: duration uncertainty by activity with three-point estimates and a stated distribution, discrete risk events mapped from the risk register, and a run configuration that makes the result reproducible. Every input records who gave it, when, by what method and what would have to be true for the extremes — because an unattributed number cannot be challenged, and an unchallenged number is where anchoring hides.

— About this document

Document	TPL-11
Series	Free Templates
Version	1.0
Status	draft
Date	2026-08-04
Unresolved placeholders	0

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Quantitative schedule risk analysis input sheet

In one paragraph. The three sheets a quantitative schedule risk analysis actually needs: duration uncertainty by activity with three-point estimates and a stated distribution, discrete risk events mapped from the risk register, and a run configuration that makes the result reproducible. Every input records who gave it, when, by what method and what would have to be true for the extremes — because an unattributed number cannot be challenged, and an unchallenged number is where anchoring hides.

Who this is for. Planners, schedulers, risk analysts and project controls managers who run or commission a quantitative schedule risk analysis; and the project managers who have to act on a P80 date they did not produce.

— 1. When to use this

A quantitative schedule risk analysis — from here, QSRA — samples uncertain activity durations and discrete risk events many times over the schedule network, and reports the distribution of completion dates that results. This sheet is where its inputs are captured, argued over and recorded.

Use it at four points. **At sanction**, to state the confidence attached to the date being committed to. **At baseline**, to size the schedule contingency and to identify which activities deserve the effort of tighter control. **At a major re-plan**, because a re-plan changes the driving path and therefore changes which uncertainties matter. **Before a completion date is contractually committed**, which is the one occasion where the difference between the deterministic date and the P80 date has a price attached.

Do not run one on a schedule that has not passed a quality review. A simulation propagates whatever logic it is given: with open ends, the uncertainty on an activity with no successor cannot reach the completion milestone, and the model will report a reassuringly tight distribution because most of the network is not connected to the answer. Run [TPL-14 – Schedule quality review checklist](#) first and record the outcome in the run configuration.

A QSRA is not a forecast of what will happen. It is a statement about the schedule as modelled, under the uncertainty as elicited. Both halves of that sentence belong in the report.

— 2. How to complete it

2.1 Decide what is in scope before you elicit anything

Model the driving paths and everything that could become one. A common and defensible approach is to model every activity with any float below an agreed threshold, plus every long-duration activity regardless of float, plus every activity a risk event attaches to. State the threshold and the reason for it in the run configuration. Modelling every activity in a ten-thousand-line schedule

is usually a way of spending three weeks to produce a less trustworthy answer than modelling four hundred of them well.

For activities in progress at the data date, model the **remaining** duration, not the original. Column H does this automatically from the status in column E.

2.2 Elicit the three-point estimate in a way that does not manufacture the answer

Anchoring is the dominant failure mode of duration elicitation, and it is not a character flaw in the estimator. Show someone a number and their subsequent estimates cluster around it. The deterministic duration in the schedule is a number, and it is usually on the screen. Six practices counter it, and they cost nothing but discipline.

Hide the deterministic duration during the elicitation. If the estimator can see 40 days, the answers will orbit 40 days. Ask for the three-point estimate cold, and only then reveal the schedule duration and discuss the difference. Column X flags every row where the most likely equals the schedule duration exactly, which is the signature of an anchored input.

Ask for the pessimistic value first, then the optimistic, then the most likely. Asking for the most likely first plants the anchor yourself. Starting at the extremes produces wider and better-calibrated ranges.

Define the extremes as frequencies, not adjectives. "Worst case" has no upper bound, so people either give an unusable number or an embarrassed one. Ask instead: if this activity were run ten times under the conditions we expect, what duration would you beat in nine of them? That is a question an experienced person can answer. Do the same at the optimistic end.

Require a written answer to "what would have to be true". Columns M and N are mandatory. An extreme with no story behind it is a number, not an estimate, and it cannot be reviewed by anyone else. This single requirement removes more bad inputs than any amount of statistical checking.

Elicit individually before the group converges. Ask each estimator separately and record the spread of answers before any discussion. Where individual answers differ by a factor of two, that disagreement is the most useful information in the whole exercise, and a workshop will destroy it in four minutes.

Name the person, not "the team". Column O takes a name and a role. Attribution is not about blame; it is about being able to go back and ask a specific person a specific question when the model produces something surprising.

Two further traps. If a discrete risk from the register is being modelled as an event on sheet 2, it must not also be baked into the pessimistic duration on sheet 1 — column L on sheet 2 flags candidates for that double count. And if the estimator's most likely duration is materially longer than the schedule duration, column AA will show it: that is not a risk finding, it is a schedule finding, and it should be dealt with as one before the simulation runs.

2.3 Choose the distribution deliberately

Three shapes cover almost everything, and the choice affects the answer.

Triangular takes the three points literally and gives real weight to the extremes. It is the honest default when the estimator has genuine views about how bad things can get, and it produces a wider spread than the alternatives.

BetaPERT weights the most likely value four times as heavily as the extremes, producing a smoother, tighter distribution. It suits activities where the most likely value is well evidenced and the extremes are softer judgements.

Uniform says every duration between the optimistic and pessimistic values is equally likely. That is a strong statement of ignorance and is occasionally the correct one — a permit determination with a statutory window and no other information, for instance. It should be rare, and it should be justified in column AE.

Record the choice per activity in column L. A model in which every activity carries the same distribution because that was the software default is not a modelling decision; it is an absence of one.

2.4 Set correlation, and record why

Activities that share a crew, a supplier, a design package, a weather window or a permitting authority do not vary independently. If the crew is slow on one, it is slow on the others. Modelling them as independent lets the simulation cancel the variation out, which narrows the distribution and produces a P80 that sits implausibly close to the P50 — the classic symptom of an uncorrelated model.

Assign a correlation group in column AB and a coefficient in column AC. Assigning correlation is judgement, not measurement, and the honest treatment is to record the reason for each group in column AE, run the model at the assigned coefficients, then re-run at a materially different set and report how much the answer moved. If it moves a lot, correlation is the dominant assumption in your analysis and the report should say so.

— 3. The template

Three sheets. Headers in row 1, data from row 2, formulas written for row 2 and filled down.

3.1 Sheet 1 — Durations: input columns

Col	Field	What goes in it
A	Activity ID	Exactly as it appears in the schedule, so the mapping can be verified
B	Activity name	As in the schedule
C	WBS or discipline	For grouping the results
D	Calendar ID	The activity's calendar in the schedule. Durations are in that calendar's working days
E	Status	Not started · In progress · Complete
F	Deterministic duration	The original duration in the current schedule, in working days
G	Remaining duration at the data date	For in-progress activities only
I	Optimistic duration (O)	The duration that would be beaten only about one time in ten
J	Most likely duration (M)	The single most probable duration
K	Pessimistic duration (P)	The duration that would be beaten about nine times in ten
L	Distribution	Triangular · PERT · Uniform
M	Basis of the optimistic value	What would have to be true. Mandatory
N	Basis of the pessimistic value	What would have to be true. Mandatory
O	Elicitation source	Name and role of the person who gave the estimate
P	Elicitation date	
Q	Elicitation method	Individual interview · Facilitated workshop · Reference class · Analyst assumption
R	Input confidence	High · Medium · Low. Drives the review order, not the model
AB	Correlation group	A short group code, e.g. CIVILS-CREW-A
AC	Correlation coefficient	The coefficient assigned within the group, as a decimal
AD	Risk drivers mapped	Risk IDs from the register that drive this activity's uncertainty
AE	Notes and assumptions	Including the reason for the correlation group and any unusual distribution choice

3.2 Sheet 1 — Durations: calculated columns

Col	Field	Formula in words	Spreadsheet expression
H	Duration modelled	The remaining duration if the activity is in progress, otherwise the deterministic duration	=IF(\$E2="In progress",\$G2,\$F2)
S	Order check	Confirms the three points are in ascending order; blank until all three are entered	=IF(COUNT(\$I2:\$K2)<3,"",IF(AND(\$I2<=\$J2,\$J2<=\$K2),"OK","Check order"))
T	Spread ratio	Pessimistic divided by most likely; blank if the most likely is zero or empty	=IF(N(\$J2)=0,"",\$K2/\$J2)
U	Downside days	Pessimistic less most likely	=IF(COUNT(\$J2:\$K2)<2,"",\$K2-\$J2)
V	Upside days	Most likely less optimistic	=IF(COUNT(\$I2:\$J2)<2,"",\$J2-\$I2)
W	Skew ratio	Downside days divided by upside days; reported as text where no upside was modelled	=IF(OR(\$U2="",\$V2=""),"",IF(\$V2=0,"No upside modelled",\$U2/\$V2))
X	Anchoring flag	Flags rows where the most likely duration equals the duration in the schedule exactly	=IF(OR(\$H2="",\$J2=""),"",IF(\$J2=\$H2,"Most likely equals schedule – challenge",""))
Y	Distribution mean	Triangular: the three points averaged. PERT: the three points averaged with the most likely weighted four times. Uniform: the midpoint of the two extremes	=IF(\$L2="Triangular",(\$I2+\$J2+\$K2)/3,IF(\$L2="PERT",(\$I2+4*\$J2+\$K2)/6,IF(\$L2="Uniform",(\$I2+\$K2)/2,""))
Z	Distribution standard deviation	Triangular: the closed-form standard deviation of a triangular distribution. PERT: the classical approximation, the range divided by six. Uniform: the range divided by the square root of twelve	=IF(\$L2="Triangular",SQRT((\$I2^2+\$J2^2+\$K2^2-\$I2*\$J2-\$I2*\$K2-\$J2*\$K2)/18),IF(\$L2="PERT",(\$K2-\$I2)/6,IF(\$L2="Uniform",(\$K2-\$I2)/SQRT(12),""))
AA	Mean less modelled duration	The distribution mean less the duration currently in the schedule	=IF(OR(\$Y2="",\$H2=""),"", \$Y2-\$H2)

The mean and standard deviation columns are **sense checks on the inputs, not the model**. The simulation samples the distribution itself; it does not use these figures. The PERT standard deviation in column Z is the classical approximation and differs slightly from the exact beta-PERT value — it is here to let a reviewer spot an input whose spread is implausible for the work described, which it does perfectly well.

Column AA is the one to read first at a review. If the sum of column AA over the driving path is large and positive, the schedule's durations are systematically shorter than the people doing the work believe. That is a planning problem masquerading as a risk problem, and no amount of simulation will fix it.

3.3 Sheet 2 — Risk events

Discrete events from the risk register that are modelled as occurrences rather than as duration spread.

Col	Field	Input or calculated	What goes in it
A	Risk ID	Input	Matching the ID in TPL-10 – Risk register, so the two can be reconciled
B	Risk title	Input	The event, in the register's words
C	Modelled as	Input	Discrete event · New logic branch · Both
D	Probability of occurrence	Input	As a decimal. Where it differs from the register, state why in column N
E	Optimistic added duration	Input	Working days
F	Most likely added duration	Input	Working days
G	Pessimistic added duration	Input	Working days
H	Distribution	Input	Triangular · PERT · Uniform
I	Attaches to	Input	The activity ID or logic path the impact applies to
J	New logic created	Input	Describe any rework loop, re-tender or re-mobilisation the event introduces
K	State modelled	Input	Pre-response · Post-response. Which of the register's two assessments this row represents
L	Double-count check	Calculated	Flags where the same risk ID also appears against a duration input
M	Expected added duration	Calculated	Sense check only
N	Notes	Input	Including the reason for any divergence from the register

Col	Formula in words	Spreadsheet expression
L	Flags the row if this risk ID also appears in the risk-driver column of the durations sheet	=IF(COUNTIF(Durations!\$AD:\$AD,"*"&\$A2&"")>0,"Also mapped to a duration input – check for double counting","")
M	The probability multiplied by the mean of the impact distribution	=IF(OR(\$D2="", \$H2=""), "", \$D2*IF(\$H2="Triangular", (\$E2+\$F2+\$G2)/3, IF(\$H2="PERT", (\$E2+4*\$F2+\$G2)/6, IF(\$H2="Uniform", (\$E2+\$G2)/2, ""))))

Column L is a prompt, not a proof — it catches the case where the same identifier appears in both places, not the case where the same uncertainty has been described twice in different words. That one is found by reading, at the review.

Column M is a sense check with a caveat that should be printed next to it: a discrete event does not add its expected duration to anything. It either happens, in which case it adds its sampled impact, or it does not, in which case it adds nothing. The expected value is useful for ranking the events and for nothing else.

3.4 Sheet 3 — Run configuration and outputs

Reproducibility is the point of this sheet. A QSRA that cannot be re-run to the same numbers cannot be defended.

Field	What goes in it
Schedule file name and version	The exact file analysed
Data date	
Schedule quality review reference and date	The TPL-14 review this schedule passed, and any check it failed
Activities in schedule / activities modelled	Both numbers, with the in-scope rule stated
Float threshold used for scope selection	And the reason for it
Sampling method	Latin hypercube · Simple random
Iterations	With the stability check below
Random seed	Recorded so the run can be repeated exactly
Correlation matrix reference	Where the coefficients and their reasons are held
Calendars used	And any calendar overridden for the analysis, with the reason
Exclusions	What was left out of the model and why. The most-read line on this sheet
Analyst	Name and date
Reviewer	Name and date

Outputs record, one row per milestone:

Field	What goes in it
Milestone ID and name	
Deterministic date	The date in the schedule as analysed
P50 date · P80 date · P90 date	
Working days from deterministic to P80	The number the project is actually being asked to fund or absorb
Confidence at the committed date	The percentile at which the currently committed date sits
Top five drivers	By sensitivity or correlation to the milestone, named
The sentence this supports	One sentence stating what the analysis does and does not show

Stability check. Run the model twice with different random seeds and record both P80 dates. The absolute difference between them, in working days, must be smaller than the precision you intend to report — if you are going to report a date to the day, a two-day swing between runs means the iteration count is too low. Increase iterations and re-run. State both runs and the difference on this sheet rather than reporting a single number and hoping.

3.5 Pasting it into a spreadsheet

Copy each header line into cell A1 of its own sheet and split on the pipe character.

Activity ID|Activity name|WBS or discipline|Calendar ID|Status|Deterministic duration|Remaining duration|Duration modelled|Optimistic|Most likely|Pessimistic|Distribution|Basis of optimistic|Basis of pessimistic|Elicitation source|Elicitation date|Elicitation method|Input confidence|Order check|Spread ratio|Downside days|Upside days|Skew ratio|Anchoring flag|Distribution mean|Distribution SD|Mean less modelled|Correlation group|Correlation coefficient|Risk drivers mapped|Notes and assumptions

Risk ID|Risk title|Modelled as|Probability|Optimistic added|Most likely added|Pessimistic added|Distribution|Attaches to|New logic created|State modelled|Double-count check|Expected added duration|Notes

4. Worked fragment

Illustrative figures. Durations in working days on a five-day calendar. Three activities from a [Durations](#) sheet, showing the checks doing their work.

Activity	Modelled	O	M	P	Distribution	Spread ratio	Down	Up	Skew	Mean	SD	Mean less modelled	Anchor flag
A-1420 Install and terminate switchgear	20	18	22	34	PERT	1.55	12	4	3.00	23.33	2.67	+3.33	—
A-1510 Commission and energise	15	12	16	30	Triangular	1.88	14	4	3.50	19.33	3.86	+4.33	—
A-1180 Civil works to substation base	40	38	40	44	Triangular	1.10	4	2	2.00	40.67	1.25	+0.67	Most equi sche — chal

The substitutions.

A-1420, PERT mean: $(18 + 4 \times 22 + 34) \div 6 = (18 + 88 + 34) \div 6 = 140 \div 6 = 23.33$ working days. Classical PERT standard deviation: $(34 - 18) \div 6 = 16 \div 6 = 2.67$ working days. Spread ratio: $34 \div 22 = 1.55$. Mean less modelled duration: $23.33 - 20 = +3.33$ working days.

A-1510, triangular mean: $(12 + 16 + 30) \div 3 = 58 \div 3 = 19.33$ working days. Triangular standard deviation: $\text{SQRT}((12^2 + 16^2 + 30^2 - 12 \times 16 - 12 \times 30 - 16 \times 30) \div 18) = \text{SQRT}((1,300 - 1,032) \div 18) = \text{SQRT}(14.889) = 3.86$ working days.

A-1180, triangular mean: $(38 + 40 + 44) \div 3 = 122 \div 3 = 40.67$ working days. Standard deviation: $\text{SQRT}((1,444 + 1,600 + 1,936 - 1,520 - 1,672 - 1,760) \div 18) = \text{SQRT}(28 \div 18) = \text{SQRT}(1.556) = 1.25$ working days.

What the checks found. A-1180 is the row to challenge, and two columns say so independently. The most likely duration is exactly the schedule duration, so the anchoring flag fires. The spread ratio is 1.10, meaning the estimator believes a forty-day civils activity will finish within four days of plan nine times out of ten — a claim that would be remarkable on a site with no weather, no ground and no other trades. The standard deviation of 1.25 days on a forty-day activity says the same thing in a different unit. Almost certainly the estimator was shown the schedule and asked to put a range around it. That row should be re-elicited before the model is run, hiding the deterministic duration and asking for the pessimistic value first.

Compare A-1510, which looks like a real estimate: a skew ratio of 3.5 says the estimator sees far more ways for commissioning to run long than to run short, which is what commissioning is like. The basis fields on that row should say what those ways are.

What the fragment does not show. The three activities have a combined mean-less-modelled figure of $3.33 + 4.33 + 0.67 = 8.33$ working days. That is **not** eight days of schedule slip. Uncertainty combines through the network rather than adding along a list: some of these activities may not be on the driving path, paths merge, and merge bias pushes the completion distribution later than any single path suggests. Only the simulation over the network can produce the completion distribution, which is exactly why this sheet is an input and not an answer.

— 5. Common mistakes

Running the model on an unreviewed schedule. Open ends, missing successors and constraints that override logic all suppress the propagation of uncertainty. The result is a tight, confident and meaningless distribution. This is the single most common way a QSRA misleads.

Anchored three-point estimates. Symptoms: most likely values that equal the schedule durations, spread ratios near 1, and empty basis fields. Columns M, N, T and X are all pointed at this failure because it is the one that quietly determines the answer.

Symmetric ranges. Duration risk is almost never symmetric — there are more ways for work to take longer than to finish early, and the ways it can finish early are bounded by physics and the supply chain while the ways it can run long are not. A register of symmetric triangles usually means the analyst applied a percentage rather than eliciting anything.

Percentage uncertainty applied globally. Applying a blanket "minus 10 %, plus 30 %" to every activity is fast, defensible-sounding and worthless: it says nothing the schedule did not already say, and it produces a ranking of drivers that is simply a ranking of durations.

Double counting. A risk modelled as a discrete event on sheet 2 and also written into the pessimistic duration on sheet 1 is counted twice. Column L catches the flagged case; the rest is found by reading the basis fields against the risk descriptions.

Ignoring correlation. Independent sampling of activities that share a crew, a supplier or a weather window produces a distribution that is too narrow. If your P80 is only a handful of days beyond your P50 on a multi-year project, correlation is the first place to look.

Reporting a P-value without the exclusions. "P80 completion is 14 March" is not a finding until it is followed by what was modelled, what was excluded, and which state — pre-response or post-response — the risk events were modelled in. A P80 that excludes the consent risk and a P80 that includes it are different numbers with the same name.

Treating the P80 as a target. It is a confidence statement about a model. Handing it to the delivery team as the new date converts a risk analysis into a schedule extension and guarantees that the next one will be gamed.

— 6. Adapting it

Safe to change. The distribution options, the in-scope selection rule, the correlation coding, the addition of columns for anything your tool imports. If your software takes distribution parameters rather than three points, add the parameter columns and keep the three points as the human-readable record of what was elicited.

Safe to add. A resource-uncertainty block where labour availability, not duration, is the driver. A weather block that models seasonal calendars as a distribution rather than an activity. A column recording the individual answers before group convergence, which is often the most informative data the exercise produces.

Do not change. The mandatory basis fields for both extremes; the named elicitation source and date; the recording of the seed and the iteration count; and the separation between duration uncertainty and discrete events. Each of those exists because a QSRA is an argument, and every one of them is a place where the argument is either supported or is not.

6.1 Before the model is run

- The schedule has passed a quality review, and any failed check is recorded with its effect on the analysis.
 - The in-scope rule is written down, and the count of activities modelled against activities in the schedule is stated.
 - Every modelled activity has all three points, a stated distribution, and both basis fields completed.
 - The order check reports OK on every row.
 - Every anchoring flag has been either cleared by re-elicitation or annotated with why the estimate stands.
 - Every spread ratio below an agreed floor has been reviewed and either corrected or justified.
 - In-progress activities are modelled on remaining duration, and the data date matches the schedule.
 - Every discrete risk event names a register ID, states which response state it models, and has been checked for double counting against the duration inputs.
 - Correlation groups have a written reason each, and a sensitivity run at different coefficients is either done or scheduled.
 - The seed, the iteration count and the stability check between two seeds are recorded.
 - The exclusions line is complete and has been read out loud to whoever will act on the result.
-

— Related

- [BPG-17 – Quantitative schedule risk analysis](#) — the method: what the simulation does, how merge bias arises, and how to read a tornado chart without over-reading it
- [BPG-05 – Schedule quality – a practical review](#) — why an unreviewed schedule cannot carry a simulation
- [TPL-14 – Schedule quality review checklist](#) — the review that must precede the run, and the record the run configuration references
- [TPL-10 – Risk register](#) — the source of the discrete events on sheet 2, and the register that must reconcile to them
- [BPG-10 – Contingency and management reserve](#) — how the output distribution becomes a schedule contingency someone will actually approve

— Sources and standards

This is an original instrument developed by the Institute. It reproduces no third-party template, form or worked example. The triangular, PERT and uniform distributions and their closed-form means and standard deviations are standard results in probability, computed here rather than quoted, and every figure in §4 was independently recomputed before publication. The classical PERT standard-deviation approximation is identified as an approximation wherever it is used. The elicitation guidance in §2.2 describes practice in the Institute's own words and does not reproduce any published protocol.

— Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.